Four reports, two graph comparisons

Research disposition for Kirtus Justus and Claude. The four pasted texts and the F0 diagram are preserved in the accompanying evidence.

What agrees exactly

The rook graph and the certified weighted alternative both have 16 vertices. For the normalized antisymmetric source on the specified pair, $m_j = u^T(L^+)^j u$. Both laws give $(m_0,m_1,m_2,m_3) = (1, 5/32, 7/256, 11/2048)$. The rook value of m_4 is 19/16384. The alternative has $m_4 \approx 0.006804564875115$. The fourth power is the fifth entry when normalization is counted.

Why the certificate residuals differ

The retained D center gives Newton displacement 4.3861×10^{-51}. B uses additional digits in its rational center and obtains 3.6891×10^{-63}. Both contraction boxes have radius 10^{-34}. Local uniqueness and positive weights follow inside the specified box. The 16-site extension preserves its pair measure. These are checks of the same underlying witness.

A different pair of graphs is blind at every scale

For adjacent antisymmetric inputs in both unweighted rook and Shrikhande, the spectral weights at eigenvalues 4 and 8 are 1/4 and 3/4. Both densities are $16s\exp(-8s) + 192s\exp(-16s)$. The graphs are nonisomorphic. In contrast, the weighted witness above has three positive coupled rates. Its full scale distribution differs from the rook edge by total variation ≈ 0.104957403. The earlier 181-photon experiment compared a rook edge with a Shrikhande nonedge using an equal-phase input.

An additional observable resolves the unweighted ambiguity

Select four labeled sites known to form a clique under the rook hypothesis. Probe all six antisymmetric pairs, or choose one uniformly per trial. Every rook pair has $m_1 = 5/32$. Shrikhande has no four-clique, so at least one of the six gives 3/16. The mean m_1 gap is at least 1/192. Enumeration gives eight four-cliques in rook and zero in Shrikhande. This control assumes access to the chosen sites and to preparations spanning them.

Arbitrarily many hidden modes

For an m-site positive complete core, attach k sites with common cross-weight ε. Subtract kε/m from every core edge, choosing ε small enough to preserve positivity. The off-diagonal cross block annihilates old zero-sum inputs and the added old block is proportional to the all-ones matrix. Thus every spectral function agrees on that whole sector, with zero outside it. Monitoring new ports gives zero signal there. Independently counting physical ports can still reveal their number.

A certified fixed-bit experiment

Two fully specified hypotheses, repeated independent preparations, an ideal positive-scale detector and a known scale unit.

Predeclare one event per photon

Return one if s belongs to (0, 0.04), (0.19, 0.55) or (2.72, ∞). Round decimal endpoints are fixed in advance. Integrating each gamma component via its survival law (1+2λs) exp(−2λs) gives probabilities 0.3768470452 for rook and 0.4817828807 for the weighted alternative. These differ from C’s exact-crossing event probabilities, respectively 0.3788905463 and 0.4838479491. Its four crossing locations are numerical.

A finite-sample crossing with both errors controlled

For equal priors the minimum error using this bit is half the sum over counts k of the smaller binomial mass. At 145 trials the computed enclosure is [0.10018508687, 0.10018508726]. At 146 it is [0.09910953485, 0.09910953520]. Ignoring a final trial makes optimal risk nonincreasing. Hence 146 is the first crossing for this bit. To control both individual errors, choose the alternative at counts above 63 and rook below 63. At exactly 63 choose rook with probability 2/25. The respective error bounds are 0.099248771 and 0.099369108. The deterministic equal-prior rule need not equalize these errors.

How the enclosure was obtained

The certificate bounds each eigenvalue and weight by rational endpoints. Survival is decreasing in λ. For each exponential, divide its argument by 256, use consecutive alternating Taylor sums through degrees 20 and 21, then square eight times with outward rounding. Bound each binomial mass by products of lower or upper endpoints, then sum. Exact arithmetic is used throughout this verifier. For the full density difference, division by 4s gives a sum of five exponentials with distinct rates. Such a sum has at most four positive zeros: factor the slowest exponential, differentiate and apply Rolle’s theorem inductively. Certified signs +, −, +, −, + at 0, 0.1, 0.3, 1 and 3 exhaust the possible crossings.

Why the other counts agree

B’s upper-tail bit at 0.028 first crosses 10% equal-prior risk at 154 trials. At 0.03 the analogous crossing is 157, consistent with the independent report’s looser 307-trial Bhattacharyya guarantee. The rounded multiband event also certifies the 419-trial Hoeffding guarantee for each error. B’s 818,981,022 trials refer to a sufficient Cantelli bound on a fourth-moment estimator for this graph pair. The 57,291,466 figure refers to a different positive-measure example.

What the rare mode contributes

For $q(z) = (z-1/8)(z-1/4)$, matching m_0 through m_3 gives $m_4 - 19/16384 = \int q^2 \, d\nu$. About 98.58% of the excess comes from the slow mode, which carries weight only 3.06×10^{-7}. Moderate scales therefore distinguish the graphs much more readily than the large raw moment suggests.

Calibration and phase information

The detector response, source purity and phase reference determine which parameters can be estimated.

Three complementary scalar checks

Set $T = \kappa S^p$ for $\kappa,p > 0$. On one positive mode S has Gamma shape 2 and rate 2λ. With complete, unbiased positive-scale records, $\mathrm{Var}(\log T) = p^2[\pi^2/6-1]$. The raw three-moment ratio $R = E[T]E[T^3]/E[T^2]^2$ equals $\Gamma(2+p)\Gamma(2+3p)/\Gamma(2+2p)^2$. Its Weierstrass factors are $1+x^2/(1+4x+3x^2)$, $x=p/k$, $k\geq 2$, strictly increasing in p. Thus R=4/3 selects p=1 in this family.

A simpler test already uses two scalar moments

On the same normalized pure-mode law define $V = E[T^2]/E[T]^2$. Then $V = \Gamma(2+2p)/\Gamma(2+p)^2$. Its logarithmic derivative is $2[\psi(2+2p)-\psi(2+p)] > 0$ because trigamma is positive. Hence V=3/2 also selects p=1. In the ideal independent-sample model the two local exponent estimates have asymptotic variance coefficients 3/4 from V and 84/25 from R. These follow by applying the delta method to moments through order six. B's 28/27 is the coefficient for R itself. None of these coefficients is a finite-sample error bar.

Source mixtures and detector loss matter

For a mixed positive spectral law, $S = X/(2\Lambda)$ with independent X of Gamma shape 2 and rate 1. The log-variance gains $\mathrm{Var}(\log \Lambda)$. V gains the multiplicative factor $E[\Lambda^{-2p}]/E[\Lambda^{-p}]^2 \geq 1$. R gains $E[\Lambda^{-p}]E[\Lambda^{-3p}]/E[\Lambda^{-2p}]^2 \geq 1$ by Cauchy–Schwarz. These identities give upper bounds on p from each observed statistic under the model. The calibrated rook-edge antisymmetric source already gives R=220/147. Scale-dependent selection and truncation can invalidate all three tests. Contrast matrices of raw moments with the same scalar factor instead obey $3M_1M_3 = 4M_2^2$ on the ideal positive sector. Moment-order gains must be characterized.

The phase advantage depends on the phase reference

For $\psi(s) = 2\lambda\sqrt{s}\exp(-\lambda s + ib\lambda s)$ the pure-state Fisher matrix has entries $2(1+b^2)/\lambda^2$, $2b/\lambda$ and 2. The phase depends only on bλ. When b is a nuisance, eliminating its tangent gives $2/\lambda^2$, attained by direct scale counting for λ. For fixed known b the single-parameter quantum information is higher if $b\neq 0$. This scalar statement does not assert joint attainability of the matrix bound. The three-port optimum and full-register results in the previous disposition retain their respective readout assumptions.

What was replayed

The retrieved B certificate, B operational verifier, independent moment audit, C frontier verifier and C wildcard checks reproduced their five retained JSON datasets in decoded values. The two pair-contrast ranks are 119; the common direction has zero image. D did not rerun the 69,696 bootstrap refits here. The stored 4/32 scale-loss rejections only characterize that synthetic design.

Fictional cosmology and its tests

The attached diagram proposes F0 as an explicit construction. Its mathematical consequences permit several different kinds of verdict.

Where the conditional deduction begins

Given a connected nonnegative graph Laplacian, $W_s = sL\exp(-sL)$. The effects $4W_s^2\,ds/s$ integrate to $I-P_0$. Adding P_0 completes the mathematical measurement. A source, physical device and interpretation as outcome probabilities are additional inputs. No physical time enters this calculation.

Quantum kinematics and the proposed clock

Functions of a single self-adjoint L commute. Their algebra alone cannot yield a noncommuting canonical pair. Finite matrices also obey $\mathrm{Tr}[Q,P]=0$. One can stipulate infinite-dimensional Fock space; that move supplies kinematics. As a proposed propagator at $\tau=0$, W at s_0 annihilates constants and has norm at most 1/e. It fails the identity requirement even on the positive sector. This rules out this specific clock dynamics. Relabeling a scale by $\log(s/s_0)$ is still mathematically permissible. A tracial state has trivial modular evolution; thermal-time constructions need a specified state and algebra.

Length and the missing cycle data

For the unit cubic lattice, the large-separation resistance equals $2\int_0^\infty [\exp(-2t)I_0(2t)]^3\,dt \approx 0.5054620197$. The integrand comes from the return heat kernel. This rules out resistance proportional to ordinary unbounded length on that example. A positive triangle and a triangle with one negative coupling have opposite gauge-invariant cycle products yet identical $|J|^2$ and identical L. No measurement determined solely by this L can recover which product was present. Even perfect graph reconstruction leaves this ambiguity.

The bottom box requires three verdicts

Cycle signs or phases are discarded by F0.2. A thermodynamic arrow requires dynamics, a state and boundary conditions. Gravity and cosmic expansion have no specified evolution law or quantitative observable here. Their empirical predictions remain undefined. Failure of the clock and linear-length proposals invalidates F0’s advertised joint derivation. It does not constitute an observational exclusion of every conceivable completion.

A falsification condition for F1

If a functional depends only on the adjacent-pair scale law, it takes equal values on rook and Shrikhande. Adding penalties based solely on vertex count, degree, edge count or scalar spectral traces still ties them. A regularizer can select one using additional structure; the choice then needs an independent justification. Specify the equivalence relation, interventions, functional and stability criterion. A unique minimum alone does not prove stability of a physical evolution. F1 remains a fictional target.

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What follows from the fictional postulates?

Conditional mathematics, failed implications and still-undefined dynamics in the supplied diagram

F0.1 / FICTIONAL INPUT

Signed relations

Vertices and signed or complex couplings J are supplied. No space, clock or quantum algebra has been selected.

→

F0.2 / STIPULATED MAP

Graph generator

$L = B\,\mathrm{diag}(|J|^2)\,B^T$. Positivity is exact. Connectivity is assumed. The map erases signs and phases.

→

EXACT / GIVEN L

Spectral wavelets

$W_s = sL\exp(-sL)$. The positive-scale effects integrate to $I - P_0$. Including P_0 normalizes the measurement.

Claim tested

F0.3 / REQUIRES A POSTULATE

Quantum matter

Functions of L commute. Finite matrices cannot realize $[Q,P] = iI$. A Fock space introduces new quantum structure.

F0.5 / PROPAGATOR FAILS

Log-scale clock

W at the proposed time origin kills the constant mode. The filters also fail the group law. A physical clock requires dynamics.

F0.4 / METRIC FAILS TEST

Resistance as length

Resistance approaches 0.505462 for distant sites on the unit cubic lattice. Ordinary coordinate distance grows without bound.

F0.6–F0.8 / DISTINCT DISPOSITIONS

Cycle signs and cosmic history

Opposite cycle-sign products can give exactly the same L. Arrow, gravity and expansion require specified dynamics, initial conditions and observables. The diagram supplies no evolution to test those predictions.

F1 / OPEN FICTIONAL TARGET

Operational selection

Rook and Shrikhande agree for a single adjacent-pair scale record. A score built solely from that record must tie them. A unique minimizer needs richer interventions or an explicit additional selection rule.

A physical interpretation of the scale measurement remains an independent experimental input.